FLYBOAT

Flyboat Electronic LLC — North American sales & engineering center

32D Series Metal Oxide Varistors

Epoxy-encapsulated zinc oxide varistors · Datasheet

The 32D series is a line of 32 mm round, epoxy-encapsulated zinc oxide varistors made by Guiyang High-Tech Yige Electronic Co., Ltd. for surge protective device (SPD) manufacturers and OEMs. Rated In 20 kA / Imax 40 kA (8/20 µs), the standard range covers varistor voltages from 100 V to 1,200 V in 27 model grades. The varistor discs are pressed, sintered, metallized, coated and tested in our own Guiyang plant — not bought in — which is how tolerances and surge ratings stay consistent from batch to batch. In North America, the series is supported by Flyboat Electronic LLC, the factory's sales and engineering center in Bellevue, WA.

Features

- Centre melt-through under power-frequency overload: the melt-through (puncture) point is engineered to occur at the centre of the disc, underneath the copper electrode, so heat is absorbed and conducted straight to the disconnection point — the chip resists melt-through longer and a mating thermal disconnector trips faster.
- For top grades the melt-through point is confined within a 0.5 cm radius of the chip centre, converting a random failure location into a controlled one.
- If the chip does melt through, the flame jet is shielded by the copper electrode, which conducts the heat to the trip point and drives the thermal disconnector open.
- Exceptional damp-heat resistance: under a customer's severe boil test (30 min in boiling water, cool, measure, re-boil; 6 cycles required with no significant change in varistor voltage or leakage), Yige product has reached 8 cycles with small-current parameters essentially unchanged.
- Long, verifiable service life (Arrhenius law): verified minimum expected life of 110 years at 40 °C, 11 years at 65 °C, 0.7 years at 95 °C upper ambient temperature (115 °C / 1,000 h ageing basis).
- High TOV withstand: at a temporary overvoltage of 1.45 × Uc the chip stays thermally stable for more than 45 min (r.m.s.), with no damage.

Certification status

The 32D series is not UL recognized: the catalog ratings tables for this series print the UL / C-UL / TÜV columns but leave every cell blank. For series not yet UL recognized, we can pursue UL certification for your program.
For series not yet UL recognized, we can pursue UL certification for your program. Full certification coverage for all series is on the Certifications page at www.flyboatmov.com.

Model naming: 32D K 431 — 32 = nominal size (mm); K = varistor-voltage tolerance ±10 % (J = ±5 %); 431 = varistor-voltage code (first two digits significant, last digit = number of zeros; 431 = 430 V). Further suffixes give the lead / structure form.

U1mA = varistor voltage at DC 1 mA; ΔU1mA = tolerance; UAC / UDC = maximum continuous operating voltage; In = nominal discharge current (8/20 µs); Imax = maximum discharge current (8/20 µs); UP = clamping voltage at In (reference value–ratio); ETM = maximum energy absorption (2 ms).

32D standard ratings — full table

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
32DK101	100	±10	60	85	20	40	360-3.6	150
32DK121	120	±10	75	100	20	40	400-3.6	185
32DK151	150	±10	95	125	20	40	450-3.0	230
32DK181	180	±10	115	150	20	40	490-2.8	270
32DK201	200	±10	130	170	20	40	520-2.6	310
32DK221	220	±10	140	180	20	40	550-2.6	350
32DK241	240	±10	150	200	20	40	600-2.6	370
32DK271	270	±10	175	225	20	40	680-2.6	430
32DK301	300	±10	195	250	20	40	770-2.6	480
32DK331	330	±10	210	270	20	40	830-2.55	520
32DK361	360	±10	230	300	20	40	910-2.55	580
32DK391	390	±10	250	320	20	40	1050-2.55	640
32DK431	430	±10	275	350	20	40	1150-2.55	730
32DK471	470	±10	300	385	20	40	1250-2.55	780
32DK511	510	±10	320	410	20	40	1300-2.55	840
32DK561	560	±10	350	450	20	40	1400-2.5	910
32DK621	620	±10	385	505	20	40	1550-2.5	990
32DK681	680	±10	420	560	20	40	1700-2.5	1000
32DK711	710	±10	440	585	20	40	1750-2.45	1010
32DK751	750	±10	460	615	20	40	1800-2.45	1020
32DK781	780	±10	485	640	20	40	1950-2.45	1030
32DK821	820	±10	510	670	20	40	2000-2.40	1080
32DK911	910	±10	550	745	20	40	2100-2.3	1170
32DK951	950	±10	575	765	20	40	2200-2.3	1180
32DK102	1000	±10	625	825	20	40	2400-2.3	1220
32DK112	1100	±10	680	895	20	40	2700-2.3	1350
32DK122	1200	±10	750	1060	20	40	3000-2.3	1400

32D low-residual-voltage ratings

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
32DK431	430	±10	275	350	20	40	950-2.2	730

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
32DK471	470	±10	300	385	20	40	1030-2.2	780
32DK511	510	±10	320	410	20	40	1130-2.2	840
32DK561	560	±10	350	450	20	40	1240-2.2	910
32DK621	620	±10	385	505	20	40	1350-2.15	990
32DK681	680	±10	420	560	20	40	1480-2.15	1000
32DK711	710	±10	440	585	20	40	1540-2.15	1010
32DK751	750	±10	460	615	20	40	1630-2.15	1020
32DK781	780	±10	485	640	20	40	1670-2.15	1030
32DK821	820	±10	510	670	20	40	1850-2.1	1080
32DK911	910	±10	550	745	20	40	1940-2.1	1170
32DK951	950	±10	575	765	20	40	2040-2.1	1180
32DK102	1000	±10	625	825	20	40	2200-2.1	1220
32DK112	1100	±10	680	895	20	40	2380-2.1	1350
32DK122	1200	±10	750	1060	20	40	2620-2.1	1400

Ratings as published at www.flyboatmov.com/32d-series.html.

Samples and design-in support — North America: Thomas Dong, Flyboat Electronic LLC (thomasd@flyboatmov.com); China: Colin Fei (colinf@flyboatmov.com).